# From folder to finished report: a three-day walkthrough

Consumer Decision-Making Analysis in R

The Science Repository — workshop edition

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# 1 Why this document exists

This is the **one orchestration file** for the workshop. It walks you through the analysis end-to-end and shows how the folder structure you built on Day 1 removes duplication on Days 2 and 3.

The rule for this document: **nothing is redefined here**. Variable names, cleaning logic, models, plots — they all live in the `R/` folder. This walkthrough just `source()`s them and tells the story.

That is the whole point. Change a model in `R/functions/analysis.R` and the website, the PDF, the manuscript, and the Shiny app all update the next time you render. No copy-paste.

## 1.1 How to render

One source file, three formats, three commands:

```
quarto render reports/00_walkthrough.qmd --to html   # -> docs/reports/00_walkthrough.html
quarto render reports/00_walkthrough.qmd --to pdf    # -> manuscript/output/walkthrough.pdf
quarto render reports/00_walkthrough.qmd --to docx   # -> manuscript/output/walkthrough.docx
```

Or render all three in one go: `quarto render reports/00_walkthrough.qmd`.

### 1.1.1 How does Quarto know where each format goes?

It doesn't, on its own. Quarto puts **every** rendered file under `output-dir: docs` (set in `_quarto.yml`). Routing then happens in two small steps:

1. `output-dir: docs` sends all output to `docs/`. After a render, you would see `docs/reports/00_walkthrough.html`, `docs/reports/walkthrough.pdf`, `docs/reports/walkthrough.docx`.
2. **A post-render script** (`scripts/post_render.R`) runs after every render and moves the `.pdf` and `.docx` files out of `docs/` and into `manuscript/output/`. HTML stays put.

That post-render hook is wired in `_quarto.yml`:

```yaml
project:
  type: website
  output-dir: docs
  post-render:
    - scripts/post_render.R
```

Quarto sets the environment variable `QUARTO_PROJECT_OUTPUT_FILES` to the list of files it just produced. The R script reads that variable, picks the `.pdf` and `.docx` entries, and moves them. The whole thing is ~20 lines — read it if you want to add a third destination later (e.g., `slides/` for RevealJS).

### 1.1.2 PDF prerequisite

PDF needs a LaTeX engine. One-time setup:

```r
install.packages("tinytex")
tinytex::install_tinytex()
```

If you skip this, the `--to pdf` command will fail but `--to html` and `--to docx` will keep working.

## 2 Day 1 — The scaffold

What you built yesterday is a **project**, not a folder. The difference:

```
folder_overview <- tibble::tribble(
  ~Folder,         ~Purpose,
  "R/",            "Code: numbered scripts + reusable functions.",
  "data/mock/",    "Synthetic, committed, LLM-safe data.",
  "data/raw/",     "Real data. Gitignored.",
  "data/processed/", "Output of cleaning. Reproducible, gitignored.",
  "reports/",      "Quarto sources (this file).",
  "docs/",         "Rendered website. GitHub Pages target.",
  "manuscript/",   "LaTeX paper.",
  "shiny/",        "Interactive app.",
  "references/",   "One .bib file used by both Quarto and LaTeX.",
  "figures/",      "Plots saved by scripts, shared by every output."
)
knitr::kable(folder_overview)
```

Table 1: What every folder in this project does.

| Folder | Purpose |
|---|---|
| R/ | Code: numbered scripts + reusable functions. |
| data/mock/ | Synthetic, committed, LLM-safe data. |
| data/raw/ | Real data. Gitignored. |
| data/processed/ | Output of cleaning. Reproducible, gitignored. |
| reports/ | Quarto sources (this file). |
| docs/ | Rendered website. GitHub Pages target. |
| manuscript/ | LaTeX paper. |
| shiny/ | Interactive app. |
| references/ | One .bib file used by both Quarto and LaTeX. |
| figures/ | Plots saved by scripts, shared by every output. |

You also wrote a smart `.gitignore` so real data can never be committed by accident, and an `.Renviron.example` with one switch:

```
DATA_MODE=mock   # default - read from data/mock/
DATA_MODE=real   # only when you mean it - read from data/raw/
```

Everything that follows in this document reads through `data_path()`. Flip `DATA_MODE` and the entire analysis re-runs on a different dataset, with no edits to any script.

## 3 Day 2 — From raw to results

### 3.1 Load the raw data

```
raw <- load_raw_consumer_data("consumer_data_raw.csv")
dplyr::glimpse(raw)
```

```
Rows: 300
Columns: 26
$ ParticipantID          <chr> "P001", "P002", "P003", "P004", "P005", "P006",~
$ Condition              <chr> "scarcity_frame", "combined_frame", "social_pro~
$ Age                    <dbl> 36, 63, 41, 51, 38, 46, 46, 43, 40, 27, 58, 58,~
$ Gender                 <chr> "female", "M", "Female", "non-binary", "W", "ma~
$ IncomeGroup            <chr> "low", "low", "medium", "medium", "medium", "me~
$ ProductCategory        <chr> "food", "clothing", "services", "electronics", ~
$ ps_1                   <dbl> 5, 2, 5, 5, 2, 4, 6, 4, 4, 3, 3, 4, 6, 2, 2, 5,~
$ ps_2                   <dbl> 7, 4, 5, 4, 3, 5, 6, 5, 5, 1, 3, 2, 6, 4, 2, 2,~
$ ps_3                   <dbl> 7, 3, 3, 4, 3, 4, 5, 3, 4, 2, 5, 3, 4, 5, 3, 5,~
$ sc_1                   <dbl> 3, 4, 6, 3, 4, 4, 4, 4, 4, 5, 5, 4, 6, 4, 4, 4,~
$ sc_2                   <dbl> 3, 4, 6, 2, 4, 4, 3, 4, 6, 4, 5, 4, 7, 4, 5, 3,~
$ sc_3                   <dbl> 3, 4, 5, 5, 4, 4, 4, 5, 4, 5, 5, 4, 6, 5, 6, 4,~
$ bt_1                   <dbl> 6, 4, 2, 4, NA, 3, 6, 4, 4, 3, 3, 6, 7, 2, 4, 6~
$ bt_2                   <dbl> 6, 4, 1, 3, 5, 3, 6, 4, 5, 1, 4, 4, 5, 4, 3, 5,~
$ bt_3_r                 <dbl> 2, 2, 5, 6, 1, 4, 3, 3, 5, 5, 5, 3, 1, 5, 5, 4,~
$ pv_1                   <dbl> 4, 5, 3, 3, 4, 3, 4, 3, 4, 3, 4, 3, 5, 3, 4, 5,~
$ pv_2                   <dbl> 5, 3, 5, 4, 4, 2, 3, 4, 4, 5, 4, 5, 6, 3, 5, 4,~
$ pv_3                   <dbl> 5, 4, 2, 3, 3, 3, 3, 4, 5, 3, 5, 3, 6, 4, 5, 4,~
$ perceived_quality      <dbl> 6, 3, 3, 4, 4, 3, 5, 4, 4, 3, 4, 4, 5, 2, 4, 4,~
$ perceived_social_norm  <dbl> 4, 5, 5, 4, 5, 5, 6, 4, 3, 4, 5, 4, 6, 2, 5, 5,~
$ pi_1                   <dbl> 3, 4, 4, 2, NA, 5, 4, 3, 3, 3, 2, 5, 4, 5, 5, 5~
$ pi_2                   <dbl> 2, 4, 4, 2, 3, 4, 4, 4, 2, 4, 4, 3, 4, 3, 4, 5,~
$ pi_3                   <dbl> 5, 3, 4, 4, 3, 5, 4, 3, 3, 3, 3, 5, 5, 5, 6, 5,~
$ willingness_to_pay     <dbl> 35.31, 39.74, 39.68, 44.55, 23.99, 48.11, 41.49~
$ choice                 <dbl> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 1,~
$ number_products_bought <dbl> 2, 4, 3, 1, 3, 2, 0, 2, 0, 4, 1, 2, 4, 1, 2, 3,~
```

The CSV is full of realistic mess: mixed-case column names, trailing whitespace, `-99` missing codes, inconsistent gender labels, a reverse-scored item, scattered `NA`s. See `data/mock/codebook.md` for the full inventory.

## 3.2 Clean it

The cleaning logic lives in `R/functions/data_loading.R` so the cleaning script, this walkthrough, and the Shiny app all share one implementation:

```r
clean <- clean_consumer_data(raw)
dplyr::glimpse(dplyr::select(clean, participant_id, condition, gender,
                             income_group, purchase_intention,
                             perceived_value, price_sensitivity))
```

```
Rows: 300
Columns: 7
$ participant_id     <chr> "P001", "P002", "P003", "P004", "P005", "P006", "P0~
$ condition          <fct> scarcity_frame, combined_frame, social_proof_frame,~
$ gender             <fct> woman, man, woman, non_binary, woman, man, man, wom~
$ income_group       <ord> low, low, medium, medium, medium, medium, medium, h~
$ purchase_intention <dbl> 3.333333, 3.666667, 4.000000, 2.666667, 3.000000, 4~
$ perceived_value    <dbl> 4.666667, 4.000000, 3.333333, 3.333333, 3.666667, 2~
$ price_sensitivity  <dbl> 6.333333, 3.000000, 4.333333, 4.333333, 2.666667, 4~
```

What `clean_consumer_data()` does (see source for the details): standardises column names with `janitor::clean_names()`, trims whitespace, recodes gender, turns `-99` into `NA`, reverses `bt_3_r`, and averages each item set into a scale score.

## 3.3 Descriptives

```
desc <- descriptives_by_condition(clean)
knitr::kable(desc, digits = 2)
```

Table 2: Outcomes by experimental condition.

| condition | n | mean_pi | sd_pi | mean_pv | sd_pv | mean_wtp | prop_choice_yes |
|---|---|---|---|---|---|---|---|
| control | 50 | 4.03 | 1.13 | 4.09 | 1.07 | 40.12 | 0.40 |
| scarcity_frame | 75 | 4.20 | 1.01 | 4.12 | 1.07 | 43.05 | 0.35 |
| sustainability_frame | 67 | 4.35 | 1.06 | 4.51 | 1.04 | 43.83 | 0.52 |
| social_proof_frame | 53 | 4.28 | 1.19 | 4.25 | 0.93 | 44.35 | 0.40 |
| combined_frame | 55 | 4.60 | 0.96 | 4.88 | 1.04 | 48.55 | 0.47 |

## 3.4 Visual first, model second

```
plot_pi_by_condition(clean)
```

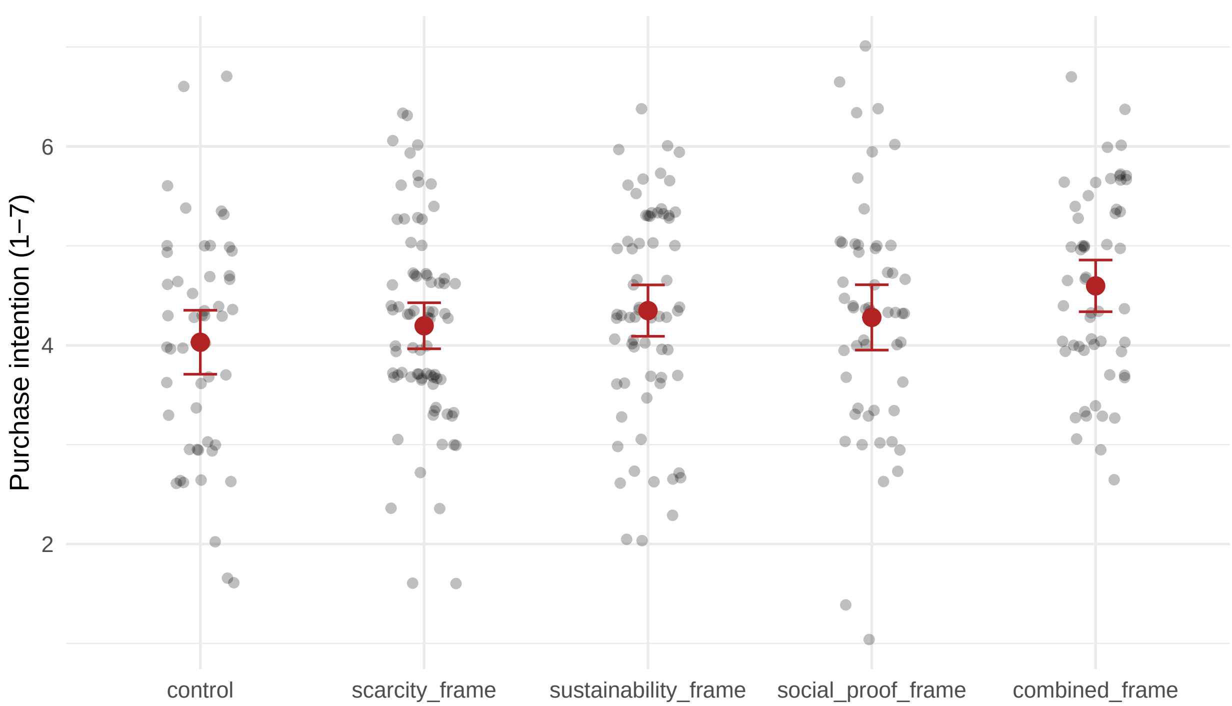


Figure 1: Purchase intention by condition. Red points are condition means; bars are 95% CIs.

`combined_frame` shows a clear bump over `control`. The scatter below suggests why: the conditions also raise `perceived_value`, and perceived value strongly predicts purchase intention.

```
plot_mediation_scatter(clean)
```

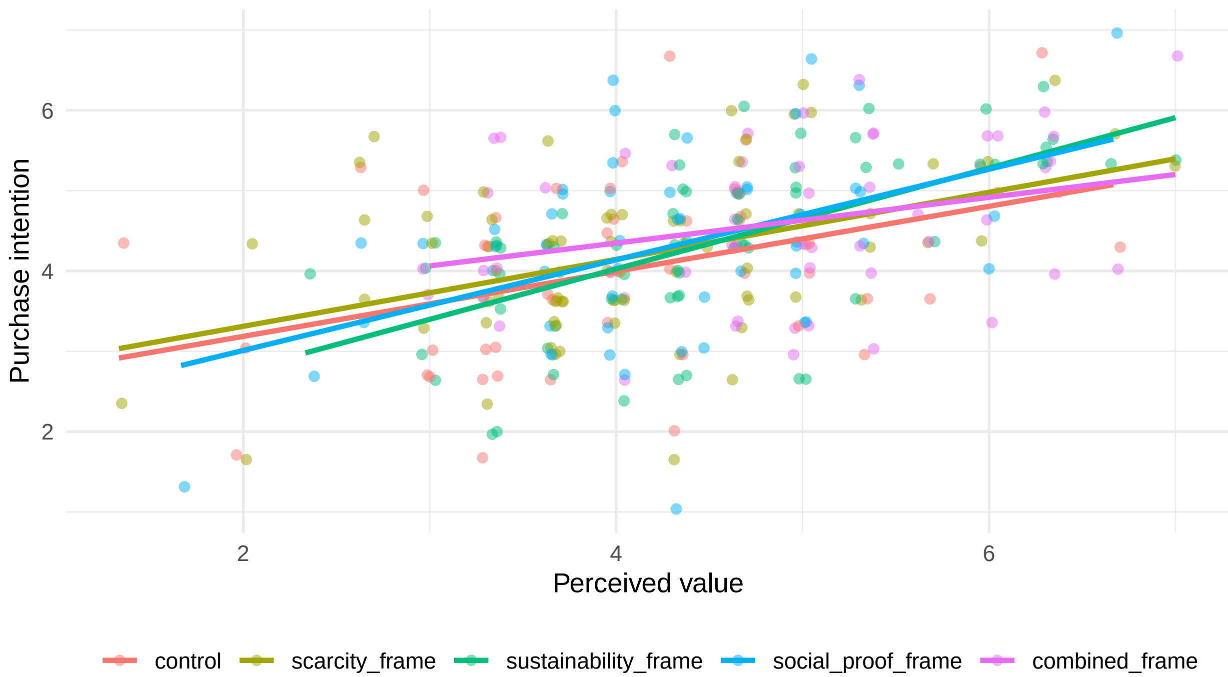


Figure 2: Perceived value (mediator) versus purchase intention (outcome), by condition.

### 3.5 Main effect: condition predicts purchase intention

```
fit_main <- fit_main_model(clean)
knitr::kable(broom::tidy(fit_main, conf.int = TRUE), digits = 3)
```

| term | estimate | std.error | statistic | p.value | conf.low | conf.high |
|---|---|---|---|---|---|---|
| (Intercept) | 4.085 | 0.254 | 16.111 | 0.000 | 3.586 | 4.584 |
| conditionscarcity_frame | 0.140 | 0.204 | 0.685 | 0.494 | -0.262 | 0.541 |
| conditionsustainability_frame | 0.311 | 0.206 | 1.508 | 0.133 | -0.095 | 0.716 |
| conditionsocial_proof_frame | 0.240 | 0.218 | 1.102 | 0.271 | -0.189 | 0.669 |
| conditioncombined_frame | 0.582 | 0.218 | 2.675 | 0.008 | 0.154 | 1.010 |
| age | -0.001 | 0.006 | -0.180 | 0.858 | -0.012 | 0.010 |
| genderman | -0.032 | 0.137 | -0.233 | 0.816 | -0.301 | 0.237 |
| gendernon_binary | -0.465 | 0.342 | -1.361 | 0.175 | -1.137 | 0.208 |
| genderprefer_not_to_say | 0.124 | 0.297 | 0.418 | 0.676 | -0.460 | 0.709 |
| income_group.L | -0.175 | 0.130 | -1.353 | 0.177 | -0.431 | 0.080 |
| income_group.Q | 0.021 | 0.104 | 0.201 | 0.841 | -0.184 | 0.226 |

### 3.6 Mediation: condition → perceived value → purchase intention

Three regressions tell the conceptual mediation story (Baron and Kenny 1986; Hayes 2017):

```r
med <- fit_mediation_models(clean)

dplyr::bind_rows(
  total       = broom::tidy(med$total, conf.int = TRUE),
  mediator    = broom::tidy(med$a,     conf.int = TRUE),
  outcome_w_m = broom::tidy(med$b,     conf.int = TRUE),
  .id = "model"
) |>
  dplyr::filter(grepl("condition|perceived_value", term)) |>
  knitr::kable(digits = 3)
```

| model | term | estimate | std.error | statistic | p.value | conf.low | conf.high |
|---|---|---|---|---|---|---|---|
| total | conditionscarcity_frame | 0.140 | 0.204 | 0.685 | 0.494 | -0.262 | 0.541 |
| total | conditionsustainability_frame | 0.311 | 0.206 | 1.508 | 0.133 | -0.095 | 0.716 |
| total | conditionsocial_proof_frame | 0.240 | 0.218 | 1.102 | 0.271 | -0.189 | 0.669 |
| total | conditioncombined_frame | 0.582 | 0.218 | 2.675 | 0.008 | 0.154 | 1.010 |
| mediator | conditionscarcity_frame | 0.045 | 0.196 | 0.227 | 0.820 | -0.342 | 0.431 |
| mediator | conditionsustainability_frame | 0.483 | 0.199 | 2.433 | 0.016 | 0.092 | 0.874 |
| mediator | conditionsocial_proof_frame | 0.196 | 0.210 | 0.934 | 0.351 | -0.217 | 0.609 |
| mediator | conditioncombined_frame | 0.849 | 0.210 | 4.050 | 0.000 | 0.436 | 1.262 |
| outcome_w_m | conditionscarcity_frame | 0.119 | 0.183 | 0.650 | 0.516 | -0.242 | 0.480 |
| outcome_w_m | conditionsustainability_frame | 0.090 | 0.187 | 0.483 | 0.630 | -0.278 | 0.459 |
| outcome_w_m | conditionsocial_proof_frame | 0.151 | 0.196 | 0.767 | 0.443 | -0.236 | 0.537 |
| outcome_w_m | conditioncombined_frame | 0.195 | 0.201 | 0.968 | 0.334 | -0.202 | 0.591 |
| outcome_w_m | perceived_value | 0.456 | 0.056 | 8.154 | 0.000 | 0.346 | 0.566 |

The pattern to read for in the table: the condition coefficients in `outcome_w_m` are **smaller** than in `total`, and `perceived_value` is a significant predictor — that's a mediation signature.

### 3.7 Moderation: price sensitivity weakens the effect

```r
fit_mod <- fit_moderation_model(clean)
knitr::kable(
  broom::tidy(fit_mod, conf.int = TRUE) |>
    dplyr::filter(grepl("condition|price", term)),
  digits = 3
)
```

| term | estimate | std.error | statistic | p.value | conf.low | conf.high |
|---|---|---|---|---|---|---|
| conditionscarcity_frame | 0.090 | 0.197 | 0.455 | 0.649 | -0.298 | 0.477 |
| conditionsustainability_frame | 0.270 | 0.199 | 1.353 | 0.177 | -0.123 | 0.662 |
| conditionsocial_proof_frame | 0.091 | 0.212 | 0.431 | 0.667 | -0.326 | 0.509 |
| conditioncombined_frame | 0.475 | 0.210 | 2.257 | 0.025 | 0.061 | 0.889 |
| price_sensitivity_c | -0.488 | 0.120 | -4.076 | 0.000 | -0.723 | -0.252 |

| term | estimate | std.error | statistic | p.value | conf.low | conf.high |
|---|---|---|---|---|---|---|
| conditionscarcity_frame:price_sensitivity_c | 0.291 | 0.155 | 1.885 | 0.060 | -0.013 | 0.596 |
| conditionsustainability_frame:price_sensitivity_c | 0.276 | 0.165 | 1.676 | 0.095 | -0.048 | 0.600 |
| conditionsocial_proof_frame:price_sensitivity_c | 0.242 | 0.164 | 1.472 | 0.142 | -0.082 | 0.565 |
| conditioncombined_frame:price_sensitivity_c | 0.354 | 0.170 | 2.076 | 0.039 | 0.018 | 0.689 |

```
plot_moderation(clean)
```

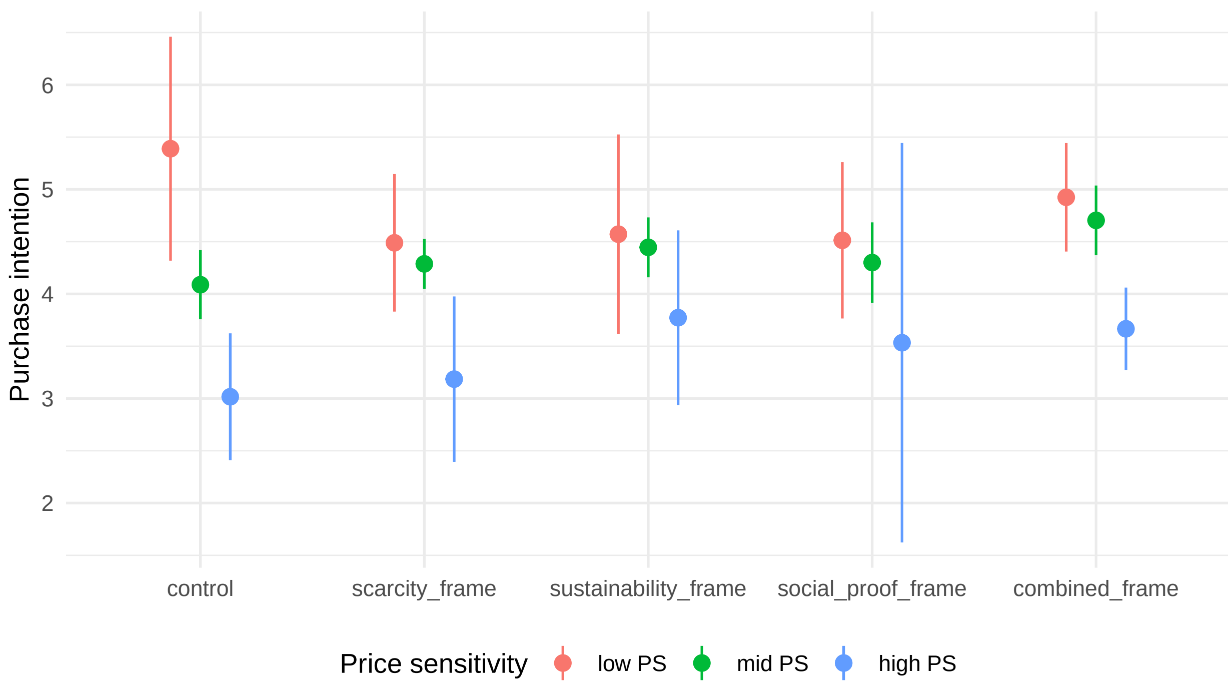


Figure 3: Condition × price sensitivity. The combined-frame boost shrinks for high-price-sensitive consumers.

# 4 Day 3 — Where does this go?

You're reading the Day 3 output right now. The same `.qmd` you're looking at becomes:

- **An HTML website** in `docs/`, served by GitHub Pages.
- **A PDF** ready to paste figures from into the LaTeX manuscript.
- **A Word document** for collaborators who don't use git.

## 4.1 The connectedness — what got reused, not copied

Every piece of this document came from somewhere else in the repo:

| Element on this page | Where it lives | Used by who else |
|---|---|---|
| Variable names (`outcome_var`, …) | `R/01_setup.R` | Every R script + Shiny app |

| Element on this page | Where it lives | Used by who else |
|---|---|---|
| `clean_consumer_data()` | `R/functions/data_loading.R` | `02_data_processing.R`, Shiny app |
| `fit_main_model()`, `fit_mediation_models()` | `R/functions/analysis.R` | `03_analysis.R`, manuscript |
| `plot_pi_by_condition()` etc. | `R/functions/plotting.R` | `03_analysis.R` (figures saved to `figures/`) |
| Citations `[@baron1986]` | `references/references.bib` | This file + the LaTeX manuscript |
| Mock data | `data/mock/` | This file + Shiny app + LLM on Day 3 |

If you renamed `purchase_intention` to `intent_to_buy`, you would change it in `R/01_setup.R` and `R/functions/data_loading.R` — **two files** — and every figure, table, PDF, website page, and the Shiny app would follow.

### 4.2 What happens when you point this at real data

Set `DATA_MODE=real` in your `.Renviron`. Re-render. Same code, same analysis, real numbers.

```
echo 'DATA_MODE=real' >> .Renviron
quarto render reports/00_walkthrough.qmd
```

The mock workflow is the rehearsal. The real workflow is one env var away.

### 4.3 On Day 3 you also brought in the LLM

If you opened this repo with the LLM-mode workspace (`the-science-repository.llm.code-workspace`), your assistant can see this file and `data/mock/` but **not** `data/raw/` or `data/processed/`. That's not a vibe; it's enforced by:

- The workspace file's `files.exclude` (editor-level hiding).
- `.claude/settings.json`'s `permissions.deny` (tool-level blocking).

So you can ask Claude to "improve the moderation plot" and it can do the job against synthetic data without ever touching the real one.

## 5 Where to take this next

- Add a new analysis: write a function in `R/functions/`, call it from `R/03_analysis.R` and from a new section here. **Don't copy code.**
- Add a new outcome: extend `descriptives_by_condition()` and re-render. All tables update.
- Write the paper: open `manuscript/main.tex`. It uses the same `figures/` and `references/references.bib`.

## 6 References

Baron, Reuben M., and David A. Kenny. 1986. "The Moderator-Mediator Variable Distinction in Social Psychological Research: Conceptual, Strategic, and Statistical Considerations." *Journal of Personality and Social Psychology* 51 (6): 1173–82. https://doi.org/10.1037/0022-3514.51.6.1173.

Hayes, Andrew F. 2017. *Introduction to Mediation, Moderation, and Conditional Process Analysis: A Regression-Based Approach*. 2nd ed. New York, NY: Guilford Press.